



FINAL MARK

GIRRAWEEEN HIGH SCHOOL
MATHEMATICS
HALF YEARLY EXAMINATION
2016
ANSWERS COVER SHEET

Name:

QUESTION	MARK	H2	H3	H4	H5	H6	H7	H8	H9
1,2	2	✓							✓
3	1	✓			✓				✓
4,5	2	✓							✓
6-10	5	✓			✓				✓
11	15	✓			✓				✓
12	15	✓			✓				✓
13ab	5	✓			✓			✓	✓
c	4	✓			✓				✓
de	6	✓			✓			✓	✓
14	15	✓			✓				✓
15a	5	✓			✓			✓	✓
b	10	✓			✓	✓			✓
16	15	✓			✓				✓
TOTALS	/100	/100			/96	/10		/16	/100

HSC Outcomes

Mathematics

- H1 seeks to apply mathematical techniques to problems in a wide range of practical contexts.
- H2 constructs arguments to prove and justify results.
- H3 manipulates algebraic expressions involving logarithmic and exponential functions.
- H4 expresses practical problems in mathematical terms based on simple given models.
- H5 applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems.
- H6 uses the derivative to determine the features of the graph of a function.
- H7 uses the features of a graph to deduce information about the derivative.
- H8 uses techniques of integration to calculate areas and volumes.
- H9 communicates using mathematical language, notation, diagrams and graphs.



GIRRAWEE HIGH SCHOOL

HALF YEARLY EXAMINATION

2016

MATHEMATICS

*Time allowed - Two hours
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Formulae are on laminated sheet provided.
- Board-approved calculators may be used.
- Answers to multiple choice questions should be coloured in on the answer sheet provided.
- Each of questions 11-16 is to be returned on a *separate* piece of paper clearly marked Question 11 , Question 12 , etc. Each piece of paper must show your name. Write END OF SOLUTIONS on your last piece of paper.
- You may ask for extra pieces of paper if you need them.

PART A: Questions 1 – 10 (Multiple Choice) *Write the letter corresponding to the correct answer on your working pages.*

Question 1

Correct to 3 significant figures, 5 745 322=

- (A) 574 (B) 575 (C) 5 740 000 (D) 5 750 000

Question 2

The gradient of a line perpendicular to $2x + 3y = 8$ is

- (A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $-\frac{3}{2}$ (D) $-\frac{2}{3}$

Question 3

$$\sum_{n=1}^7 4n - 1 =$$

- (A) 27 (B) 90 (C) 105 (D) 23

Question 4

The focus of the parabola $(x + 1)^2 = 8(y - 2)$ is

- (A) $(-1, 2)$ (B) $(1, -2)$ (C) $(-1, 0)$ (D) $(-1, 4)$

Question 5

The sum of the roots of the quadratic equation $3x^2 + 2x + 9 = 0$ is

- (A) -2 (B) $-\frac{2}{3}$ (C) 3 (D) Not real

Question 6

$$\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4} =$$

- (A) 0 (B) 4 (C) 8 (D) Undefined

Question 7

A card is drawn from a standard deck (4 cards of each kind, 13 of each suite, 52 cards altogether).
The probability that it is red or an Ace is

- (A) $\frac{1}{13}$ (B) $\frac{1}{2}$ (C) $\frac{7}{13}$ (D) $\frac{15}{26}$

Examination continues on the following page

Question 8

If $y = x^3 + 3e^{-2x}$, $\frac{dy}{dx} =$

- (A) $3x^2 - 6e^{-2x}$ (B) $3x^2 + 6e^{-2x}$ (C) $3x^2 - \frac{3}{2}e^{-2x}$ (D) $3x^2 + \frac{3}{2}e^{-2x}$

Question 9

$\int \frac{1}{(2x+1)^3} \cdot dx =$

- (A) $\frac{1}{8}(2x+1)^4 + C$ (B) $\frac{1}{8(2x+1)^2} + C$ (C) $-\frac{1}{4(2x+1)^2} + C$ (D) $\frac{1}{8(2x+1)^5} + C$

Question 10 $\int \frac{1}{e^{3x}} \cdot dx =$

- (A) $\frac{1}{2e^{2x}} + C$ (B) $\frac{1}{4e^{4x}} + C$ (C) $\frac{1}{3e^{3x}} + C$ (D) $-\frac{1}{3e^{3x}} + C$

PART B: Show all necessary working on your answer pages.

Question 11 (15 Marks)**Marks**

(a) Find $\frac{dy}{dx}$ if

(i) $y = 2\sqrt{x} - \frac{3}{x} + 2e^{3x}$

1

(ii) $y = 5x^3e^{-2x}$

3

(iii) $y = \frac{x^2}{x^3-2x+3}$

3

(iv) $y = e^{\sqrt{x}}$

2

(b) (i) Find $\int_1^3 2e^{3x} - 6x^2 \cdot dx$

3

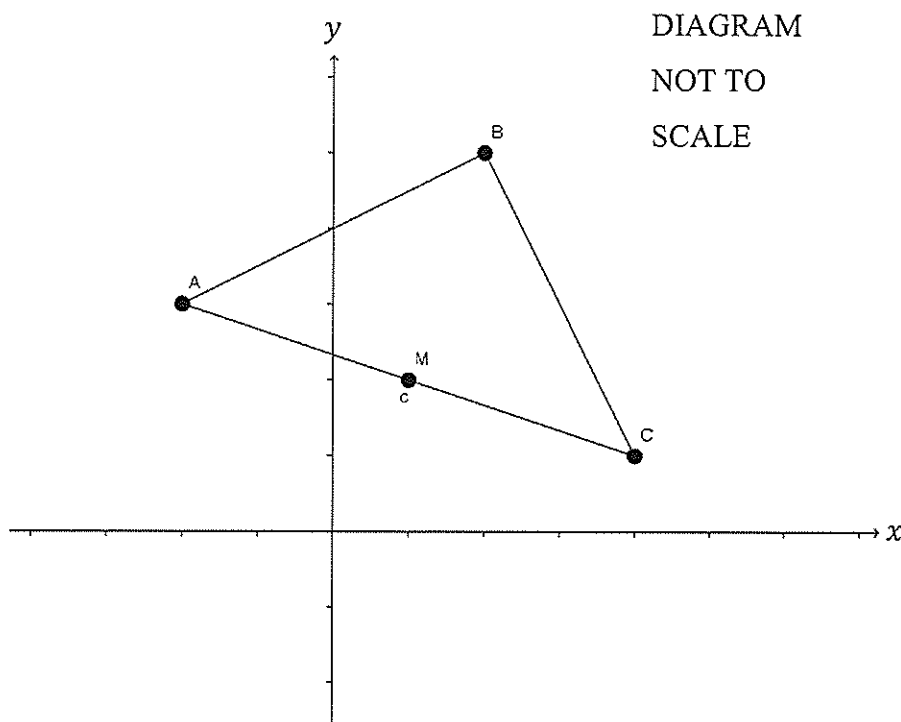
(ii) If $\frac{dy}{dx} = 3e^{2x}$ and $y = e^2$ when $x = 1$ find the expression for y .

3

Examination continues on the following page

Question 12 (15 Marks)**Marks**

- (a) $A(-2,3)$, $B(2,5)$ and $C(4,1)$ lie on the number plane. M is the midpoint of AC . (See diagram.)



- | | | |
|-------|---|---|
| (i) | Show that $AB \perp BC$ | 3 |
| (ii) | Find the coordinates of M . | 2 |
| (iii) | Show that $AM = BM = CM$. | 3 |
| (iv) | The circle which passes through A , B and C has centre M . State the equation of this circle. | 1 |
- (b) A total of 150 tickets are sold in a raffle that has 2 prizes. There are 50 red, 50 green and 50 blue tickets. At the drawing of each prize, winning tickets are NOT replaced before the next draw.
- | | | |
|-------|--|---|
| (i) | Draw a probability tree showing this information. | 2 |
| (ii) | Find the probability that at least one of the winning tickets is red. | 2 |
| (iii) | Find the probability that the winning tickets are different colours from each other. | 2 |

Examination continues on the following page

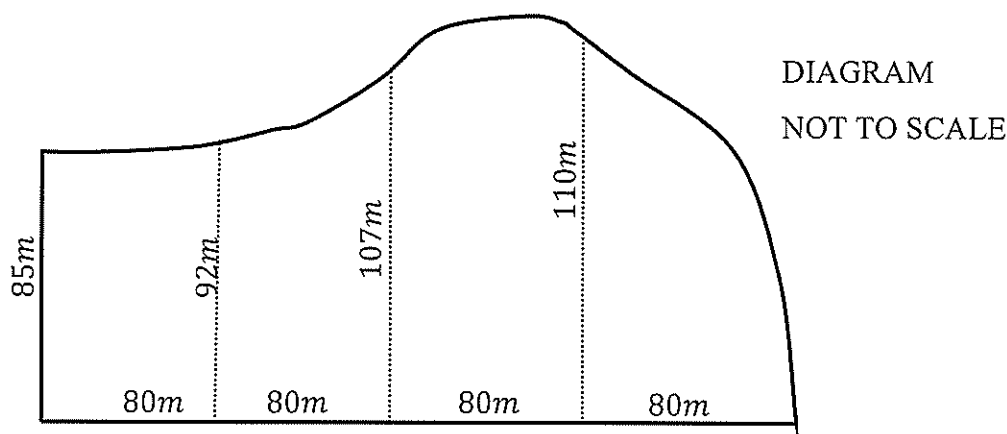
Question 13 (15 Marks)**Marks**

- (a) Use the trapezoidal with five function values to find an approximation

3

for $\int_0^1 e^{-x^2} \cdot dx$

- (b) This irregularly shaped park is bordered by a beach:

2

Use two applications of Simpson's rule to find the approximate area of the park.

- (c) (i) If $y = 5xe^x$, find $\frac{dy}{dx}$.

2

(ii) Hence find $\int 5xe^x \cdot dx$

2

- (d) Find the volume of the solid of revolution formed when $y = e^{3x}$ is rotated around the x axis between $x = -1$ and $x = 2$.

2

- (e) Find the volume of the solid of revolution formed when the area between the curves $y = 4x$, $y = \frac{1}{x}$ and the lines $y = 2$ and $y = 3$ is rotated about the y axis.

4**Question 14 (15 Marks)**

- (a) Ken wins \$500.00 and decides to keep it locked up for a rainy day.

After a while he decides to add more to his stash, so he puts in \$40 one month, then \$55, then \$70 and so on.

(Unfortunately being locked up at home it earns no interest.)

- (i) How much money will he put in in the 12th month?

1

- (ii) How much money will he have after one year?

2

- (iii) How long will it take Kevin to save \$7000?

3

Examination continues on the following page

Question 14 (continued)**Marks**

- (b) The number of people attending the Upper Kombuckta West show is increasing by 7% per year. If 3000 people attended the show in the year 2000:
- (i) How many people attended the show in 2005? 2
 - (ii) How many people attended the show altogether between 2000 and 2007? 2
- (c) Find the equation of the tangent to $y = e^{-2x}$ at the point where $x = -1$. 3
- (d) If $y = e^{2x} + 3e^{-x}$, show that $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 0$. 2

Question 15 (15 Marks)

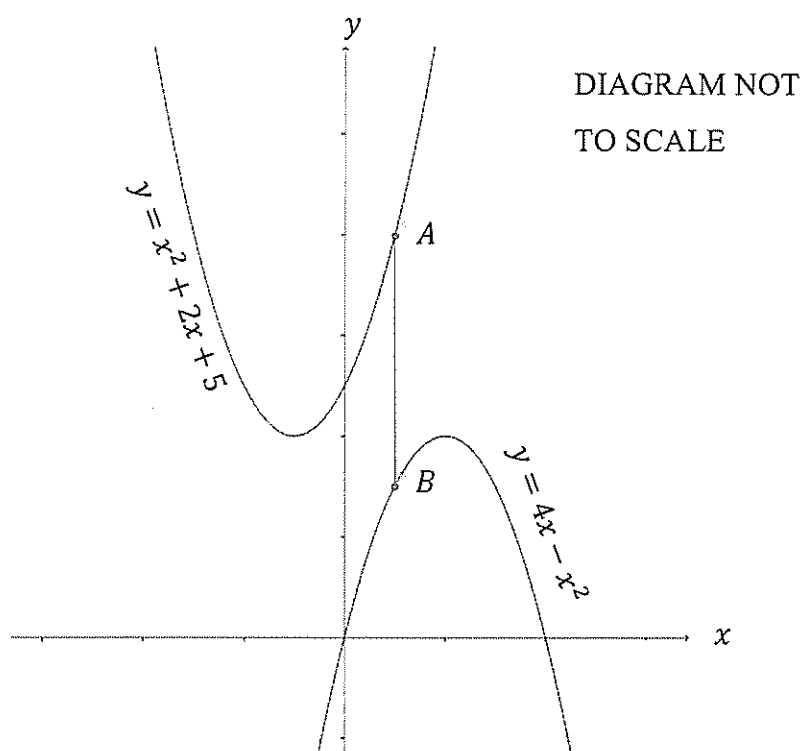
- (a) (i) Show by substitution that $y = x^2 - 1$ and $y = 3x + 3$ intersect at the points $(-1,0)$ and $(4,15)$. 2
- (ii) Find the area enclosed by $y = x^2 - 1$ and $y = 3x + 3$ 3
- (b) For the function $y = (x + 1)e^x$
- (i) Find the coordinates of the x and y intercepts. 2
 - (ii) Find any turning points and determine their nature. 3
 - (iii) Find any points of inflexion. 3
 - (iv) Sketch the graph of $y = (x + 1)e^x$ showing all intercepts, turning points and points of inflexion. 2

Examination continues on the following page

Question 16 (15 Marks)

Marks

- (a) Find the locus of the set of points that are equally distant from the point $(-1, 3)$ and the line $y = -5$. 3
- (b) A is a point on the curve $y = x^2 + 2x + 5$. B is a point on the curve $y = 4x - x^2$ with the same x coordinate as A . (see diagram.)



- (i) Using x as the x coordinate of both A and B , show that the distance $AB = 2x^2 - 2x + 5$. 2
- (ii) Find the minimum distance AB . 4
- (c) (i) Write the sum to n terms of the series $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$. 1
- (ii) For the series $1 + \frac{2}{3} + \frac{3}{9} + \frac{4}{27} + \frac{5}{81} + \dots + \frac{n}{3^{n-1}}$
- (A) Show that the sum to 5 terms can be written in the form 3

$$\frac{9}{4} \left[1 - \left(\frac{1}{3} \right)^5 \right] - \frac{5}{2} \times \left(\frac{1}{3} \right)^4$$

Hint: $1 + \frac{2}{3} + \frac{3}{9} + \dots$ can be written as $1 + \left(\frac{1}{3} + \frac{1}{3} \right) + \left(\frac{1}{9} + \frac{1}{9} + \frac{1}{9} \right) + \dots$

Examination continues on the following page

Question 16(c)(ii) (continued)

Marks

(B) Write an expression for the sum to n terms of

1

$$1 + \frac{2}{3} + \frac{3}{9} + \frac{4}{27} + \frac{5}{81} + \cdots + \frac{n}{3^{n-1}}$$

(C) Find the limiting sum of the series

1

$$1 + \frac{2}{3} + \frac{3}{9} + \frac{4}{27} + \frac{5}{81} + \cdots + \frac{n}{3^{n-1}}$$

Here endeth the examination!!!

*Remember to write END OF SOLUTIONS on your
answer page!!!*

(1) D (2) A (3) C (4) D (5) B (6) C (7) C (8) A (9) C (10) D

(11) (a) (i) $y = 2\sqrt{x} - \frac{3}{x} + 2e^{3x}$

$$= 2x^{\frac{1}{2}} - 3x^{-1} + 2e^{3x}$$

$$y' = 2 \times \frac{1}{2} x^{-\frac{1}{2}} - 3x^{-2} + 6e^{3x}$$

$$\therefore \frac{dy}{dx} = \frac{1}{\sqrt{x}} + \frac{3}{x^2} + 6e^{3x}$$

(ii) $y = 5x^3 e^{-2x}$

$$y' = u'v + v'u$$

$$= 15x^2 e^{-2x} + 5x^3 \times -2e^{-2x}$$

$$= 15x^2 e^{-2x} - 10x^3 e^{-2x}$$

$$= 5x^2 e^{-2x} (3 - 2x)$$

(iii) $y = \frac{x^2}{x^3 - 2x + 3}$

$$y' = \frac{u'v - v'u}{v^2}$$

$$= \frac{2x(x^3 - 2x + 3) - x^2(3x^2 - 2)}{(x^3 - 2x + 3)^2}$$

$$= \frac{2x^4 - 4x^2 + 6x - 3x^4 + 2x^2}{(x^3 - 2x + 3)^2}$$

$$\frac{dy}{dx} = \frac{-x^4 - 2x^2 + 6x}{(x^3 - 2x + 3)^2}$$

Q. (11)(a)(iv) $y = e^{\sqrt{x}}$

Either

By if $y = e^{f(x)} \rightarrow y' = f'(x) e^{f(x)}$

$$y' = \frac{1}{2\sqrt{x}} \times e^{\sqrt{x}}$$

$$= \frac{e^{\sqrt{x}}}{2\sqrt{x}}$$

Or

Chain rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad \left| \begin{array}{l} u = \sqrt{x} \\ \frac{du}{dx} = \frac{1}{2\sqrt{x}} \\ y = e^u \\ \frac{dy}{du} = e^u \\ = e^{\sqrt{x}} \end{array} \right.$$

$$= e^{\sqrt{x}} \times \frac{1}{2\sqrt{x}}$$

$$= \frac{e^{\sqrt{x}}}{2\sqrt{x}}$$

(b)(i) $\int_1^3 2e^{3x} - 6x^2 dx$

$$= \left[\frac{2}{3} e^{3x} - 2x^3 \right]_1^3$$

$$= \left[\frac{2}{3} \times e^{3 \times 3} - 2 \times 3^3 \right] - \left[\frac{2}{3} \times e^{3 \times 1} - 2 \times 1^3 \right]$$

$$= \frac{2}{3} e^9 - \frac{2}{3} e^3 - 52$$

$$= 5336.6656 (4DP)$$

(ii) $\frac{dy}{dx} = 3e^{2x}$

$$y = \int 3e^{2x} dx$$

$$= \frac{3}{2} e^{2x} + C$$

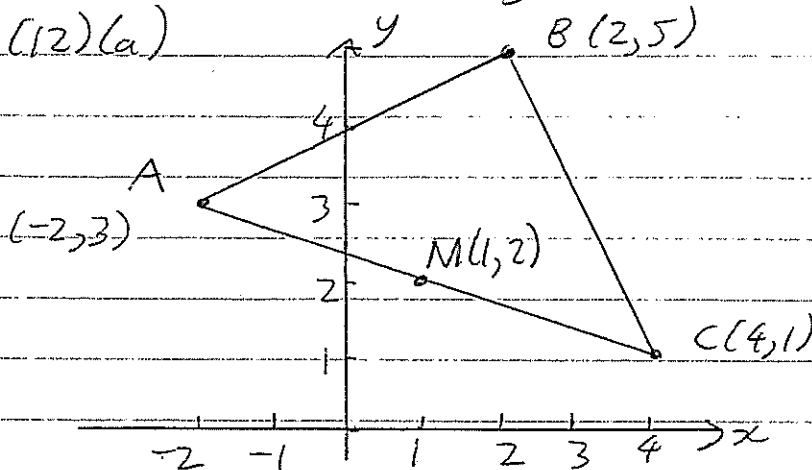
As $y = e^2$ when $x = 1$,

$$e^2 = \frac{3}{2} e^{2 \times 1} + C$$

$$-\frac{1}{2} e^2 = C$$

$$\therefore y = \frac{3}{2} e^{2x} - \frac{1}{2} e^2$$

Q. (12)(a)



$$\begin{aligned} \text{(i)} \quad m_{AB} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{5 - 3}{2 - (-2)} \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} m_{BC} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{1 - 5}{4 - 2} \\ &= -2 \end{aligned}$$

$$\begin{aligned} m_{AB} \times m_{BC} &= \frac{1}{2} \times -2 \\ &= -1 \\ \text{As } m_{AB} \times m_{BC} &= -1 \\ AB &\perp BC \end{aligned}$$

Note: * Alternative below.

$$\begin{aligned} \text{(ii)} \quad M &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &= \left(\frac{-2 + 4}{2}, \frac{3 + 1}{2} \right) \\ &= (1, 2) \end{aligned}$$

$$\begin{aligned} AC &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(4 - (-2))^2 + (1 - 3)^2} \\ &= \sqrt{40} \end{aligned}$$

Similarly

$$\begin{aligned} AB &= \sqrt{2^2 + 4^2} = \sqrt{20} \\ BC &= \sqrt{2^2 + (-4)^2} = \sqrt{20} \\ (AB)^2 + (BC)^2 &= 20 + 20 \\ &= 40 \\ &= (AC)^2 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad AM &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(1 - (-2))^2 + (2 - 3)^2} \\ &= \sqrt{10} \end{aligned}$$

$$\begin{aligned} BM &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(2 - 1)^2 + (5 - 2)^2} \\ &= \sqrt{10} \end{aligned}$$

$\therefore \Delta ABC$ is right-angled at B [Pythagoras].
 $AB \perp BC$

$$\begin{aligned} CM &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(4 - 1)^2 + (1 - 2)^2} \\ &= \sqrt{10} \\ \therefore AM &= BM = CM \end{aligned}$$

Alternative for Ext. 1 students

$$\text{As } \angle ABC = 90^\circ$$

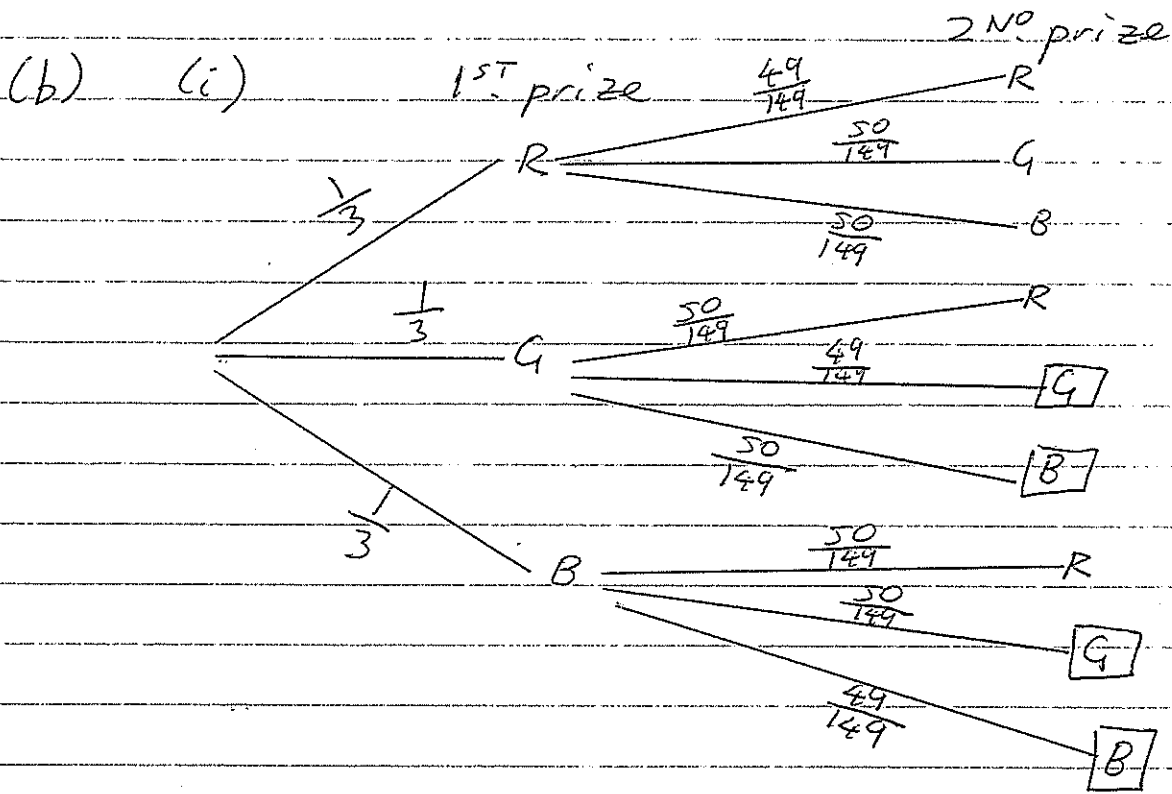
B must be at circumference of a circle with AC as diameter [Lin. in semicircle].
 As M is midpoint AC, M is midpoint of diameter & circle centre.
 $\therefore AM = BM = CM$ [circle radii].

PTO $\rightarrow$

Q. (12)(a) (continued)

(iv) Circle centre M , $r = \sqrt{10}$

$$(x-1)^2 + (y-2)^2 = 10.$$



(ii) P [at least 1 red]

$$= 1 - P(\text{Blue \& Green})$$

$$= 1 - \left[\frac{1}{3} \times \frac{49}{149} + \frac{1}{3} \times \frac{50}{149} + \frac{1}{3} \times \frac{50}{149} + \frac{1}{3} \times \frac{49}{149} \right]$$

$$= 1 - \frac{66}{149}$$

$$= \frac{83}{149}$$

$$\text{OR} = RR + RG + RB + GR + BR$$

$$= \frac{1}{3} \times \frac{49}{149} + 2 \times \frac{1}{3} \times \frac{50}{149} + 2 \times \frac{1}{3} \times \frac{50}{149}$$

$$= \frac{83}{149}$$

(iii) Winning tickets different colours

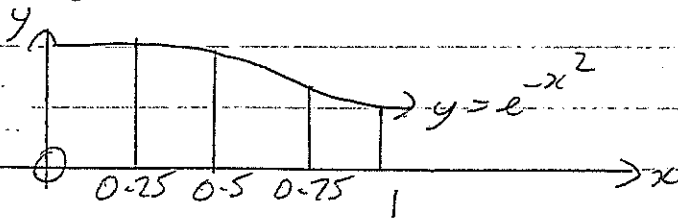
$$= 1 - (RR + GG + BB)$$

$$= 1 - 3 \times \frac{1}{3} \times \frac{49}{149}$$

$$= \frac{100}{149}$$

Y12 2U Midyear Solutions p.5

Q. (13)(a) $\int_0^1 e^{-x^2} dx \rightarrow 5$ function values
 $\rightarrow 4$ intervals of 0.25.



$$\begin{aligned} \text{By } A &\doteq \frac{h}{2} [y_0 + y_4 + 2(y_1 + y_2 + y_3)] \\ &= \frac{0.25}{2} \left[e^{-0^2} + e^{-1^2} + 2 \left(e^{-(0.25)^2} + e^{-(0.5)^2} + e^{-(0.75)^2} \right) \right] \\ &= \frac{1}{8} \left[e^0 + e^{-1} + 2 \left(e^{-\frac{1}{16}} + e^{-\frac{1}{4}} + e^{-\frac{9}{16}} \right) \right] \\ &\doteq 0.742984\dots \\ &= 0.7430 \text{ u}^2 [4 \text{ DP}] \end{aligned}$$

$$\begin{aligned} (b) \quad A &= \frac{h}{3} [y_0 + y_4 + 4(y_1 + y_3) + 2y_2] \\ &\doteq \frac{80}{3} [85 + 0 + 4(92 + 110) + 2 \times 107] \end{aligned}$$

$$A \doteq 29520 \text{ m}^2$$

$$\begin{aligned} (c) (i) \quad y &= 5xe^x \\ y' &= u'v + v'u \\ &= 5e^x + 5xe^x \end{aligned}$$

$$(ii) \text{ Hence } \int 5e^x + 5xe^x dx = 5xe^x [+ C].$$

$$\int 5e^x dx + \int 5xe^x dx = 5xe^x [+ C].$$

$$5e^x + \int 5xe^x dx = 5xe^x [+ C].$$

$$\underline{\underline{\int 5xe^x dx = 5xe^x - 5e^x + C.}}$$

Q. (13)(d) $V = \pi \int_{-1}^2 (e^{3x})^2 \cdot dx$

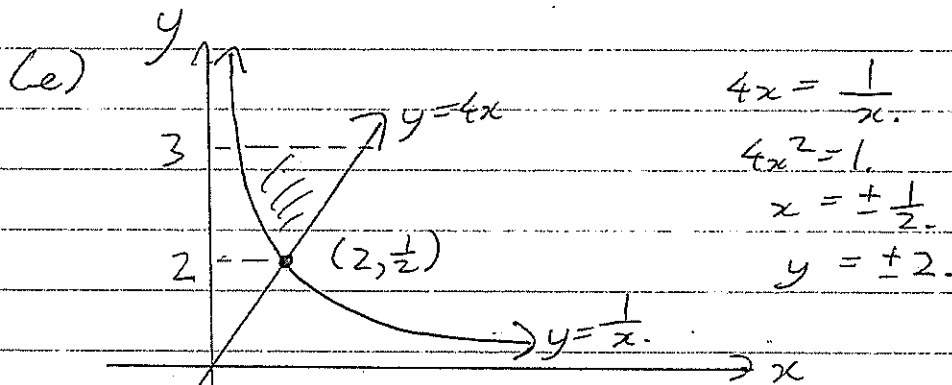
$$= \pi \int_{-1}^2 e^{6x} \cdot dx$$

$$= \pi \left[\frac{1}{6} e^{6x} \right]_{-1}^2$$

$$= \frac{\pi}{6} \left[e^{6 \times 2} - e^{6 \times -1} \right]$$

$$= \frac{\pi}{6} \left[e^{12} - \frac{1}{e^6} \right]$$

$$\approx 85\,218 \text{ u}^3 \text{ (nearest cubic unit).}$$



$$V = \pi \int_{y=2}^{y=3} (x_1)^2 - (x_2)^2 \cdot dy$$

$$\begin{aligned} y = 4x & \quad y = \frac{1}{x} \\ \frac{1}{4}y = x & \quad \frac{1}{y} = x \end{aligned}$$

$$\therefore V = \pi \int_{y=2}^{y=3} \left(\frac{1}{4}y \right)^2 - \left(\frac{1}{y} \right)^2 \cdot dy$$

$$= \pi \int_2^3 \frac{1}{16} y^2 - \frac{1}{y^2} \cdot dy$$

$$= \pi \left[\frac{y^3}{48} + \frac{1}{y} \right]_2^3$$

$$= \pi \left[\frac{3^3}{48} + \frac{1}{3} - \frac{2^3}{48} - \frac{1}{2} \right]$$

$$= \frac{11\pi}{48} \text{ u}^3 \text{ or } = 0.72 \text{ u}^3 \text{ (2DP).}$$

(i) (a) (i) 12th month $T_n = a + (n-1)d$

$$T_{12} = 40 + 11 \times 15$$

$$= \$205.$$

(ii) After 1 year: $S_n = \frac{n}{2} (a + l)$

$$= 6(40 + 205)$$

$$= \$1470.$$

But he also has \$500 so total = \$1970.

(iii) $500 + \frac{n}{2} [2 \times 40 + (n-1) \times 15] = 7000$

$$\frac{n}{2} [65 + 15n] = 6500$$

$\times \frac{2}{5}$ & expand.

$$3n^2 + 13n - 2600 = 0$$

$$n = \frac{-13 \pm \sqrt{13^2 + 4 \times 3 \times 2600}}{2 \times 3}$$

$$= 27.352..$$

→ it will take 28 months.

$$(14)(b)(i) T_n = ar^{n-1} \\ = 3000 \times 1.07^5 \\ = 4207.655..$$

→ 4208 people attended (nearest person)

(would take 4207 or 4210 (nearest 10 people))

$$(ii) GP: S_8 = \frac{a(r^n - 1)}{r - 1} \\ = \frac{3000[1.07^8 - 1]}{0.07}$$

= 30 779 people (nearest person)

or 30 780 people (nearest 10)

$$(c) y = e^{-2x}$$

$$y' = -2e^{-2x}$$

$$\text{At } x = -1, y' = -2e^{-2(-1)-1} \\ = -2e^2$$

$$\text{Also at } x = -1, y = e^{-2(-1)-1} = e^2$$

$$\text{By } y - y_1 = m(x - x_1)$$

$$y - e^2 = -2e^2(x + 1)$$

$$y - e^2 = -2e^2x - 2e^2$$

$$y = -2e^2x - e^2$$

$$\text{OR } 2e^2x + y + e^2 = 0$$

$$(d) y = e^{2x} + 3e^{-x}$$

$$y' = 2e^{2x} - 3e^{-x}$$

$$y'' = 4e^{2x} + 3e^{-x}$$

$$\therefore \frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y$$

$$= (4e^{2x} + 3e^{-x}) - (2e^{2x} - 3e^{-x}) - 2(e^{2x} + 3e^{-x})$$

$$= 4e^{2x} + 3e^{-x} - 2e^{2x} + 3e^{-x} - 2e^{2x} - 6e^{-x}$$

$$= 0 = \text{RHS.}$$

QED.

Q.15) (a) (i) When $x = -1$

$$y = x^2 - 1$$

$$= (-1)^2 - 1$$

$$= 0$$

$$y = 3x + 3$$

$$= 3 \times -1 + 3$$

$$= 0$$

When $x = 4$

$$y = x^2 - 1$$

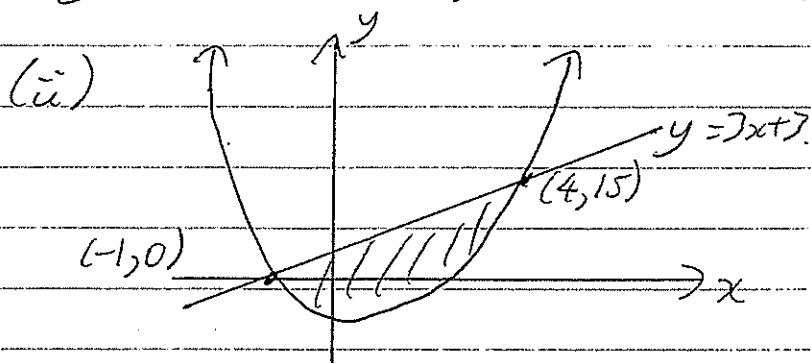
$$= 4^2 - 1$$

$$= 15$$

$$y = 3x + 3$$

$$= 3 \times 4 + 3$$

$$= 15$$

They intersect at $(-1, 0)$ & $(4, 15)$.

$$A = \int_{-1}^4 (3x + 3) - (x^2 - 1) dx$$

$$= \int_{-1}^4 3x - x^2 + 4 dx$$

$$= \left[\frac{3}{2}x^2 - \frac{1}{3}x^3 + 4x \right]_{-1}^4$$

$$= \left[\frac{3}{2} \times 4^2 - \frac{1}{3} \times 4^3 + 4 \times 4 \right] - \left[\frac{3}{2} \times (-1)^2 - \frac{1}{3} \times (-1)^3 + 4 \times -1 \right]$$

$$= \frac{56}{3} - -\frac{13}{6}$$

$$= \frac{125}{6} u^2 \text{ or } 20\frac{5}{6} u^2$$

(b) (i) $y = (x+1)e^x$

y intercept [$x=0$]

$$= (0+1) \times e^0$$

$$= \underline{1}$$

x intercept [$y=0$]

$$(x+1)e^x = 0$$

As $e^x \neq 0$ for all real x ,

$$x+1 = 0$$

$$\underline{x = -1}$$

Q.(15)(b)(ii) Turning Points:

$$\begin{aligned}
 y' &= u'v + v'u \\
 &= 1 \times e^x + e^x(x+1) \\
 &= (x+2)e^x
 \end{aligned}$$

$$y' = 0: (x+2)e^x = 0$$

$$x+2 = 0 \Rightarrow x = -2$$

$$\begin{aligned}
 \text{At } x = -2, y &= (x+1)e^x \\
 &= (-2+1)e^{-2} \\
 &= -\frac{1}{e^2}
 \end{aligned}$$

Nature of turning points

$$\begin{aligned}
 y'' &= u'v + v'u \\
 &= 1 \times e^x + e^x(x+2) \\
 &= (x+3)e^x
 \end{aligned}$$

$$\begin{aligned}
 \text{At } x = -2, y'' &= (-2+3)e^{-2} \\
 &= \frac{1}{e^2} > 0
 \end{aligned}$$

$\therefore$ MINIMUM turning point at $(-2, -\frac{1}{e^2})$

(iii) Points of inflexion:

$$\begin{aligned}
 y'' &= 0 \\
 (x+3)e^x &= 0
 \end{aligned}$$

$$x+3 = 0 \Rightarrow x = -3$$

$$\begin{aligned}
 \text{At } x = -3, y &= (x+1)e^x \\
 &= (-3+1)e^{-3} \\
 &= -\frac{2}{e^3}
 \end{aligned}$$

Testing concavity to LEFT of $x = -3$

$$\begin{aligned}
 \text{At } x = -4, y'' &= (-4+3)e^{-4} \\
 &= -\frac{1}{e^4} < 0
 \end{aligned}$$

 $\rightarrow$ Concave DOWN to left.Already tested concavity to RIGHT $\Rightarrow$ at $x = -2$

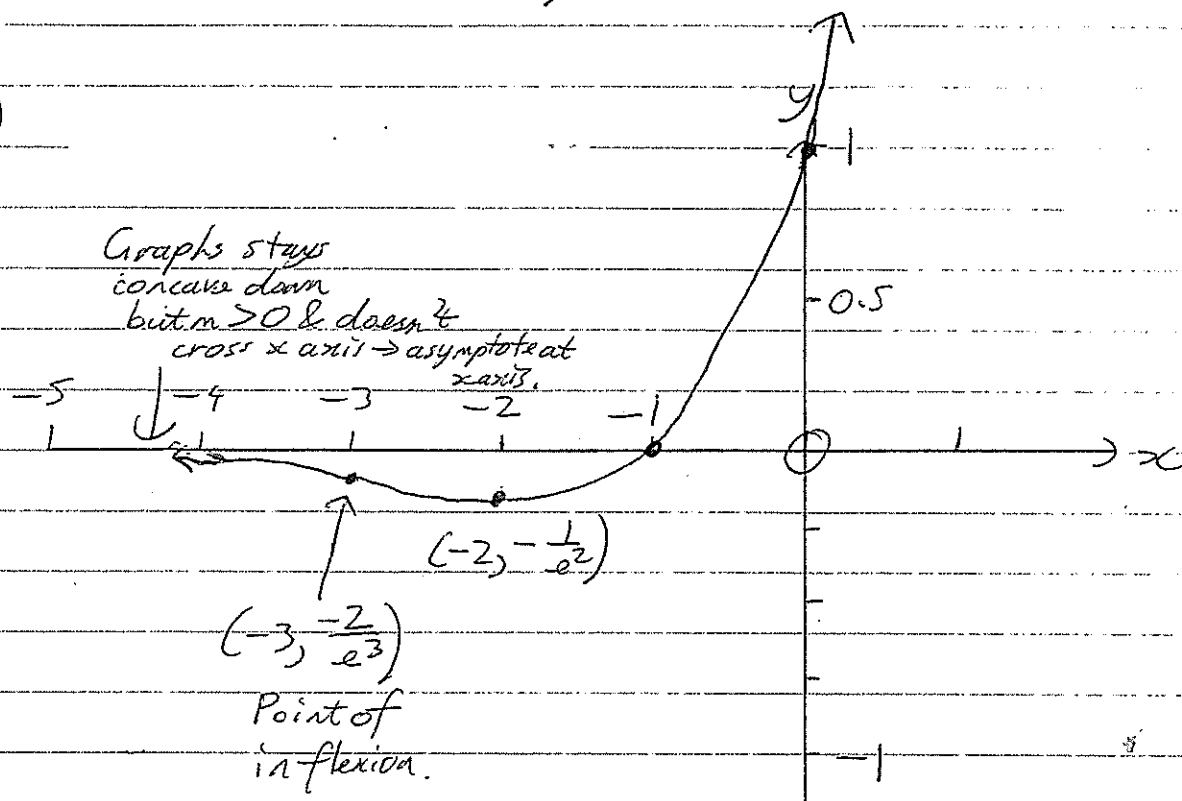
$$y'' = \frac{1}{e^2} > 0$$

Concavity CHANGES

PTO $\rightarrow$

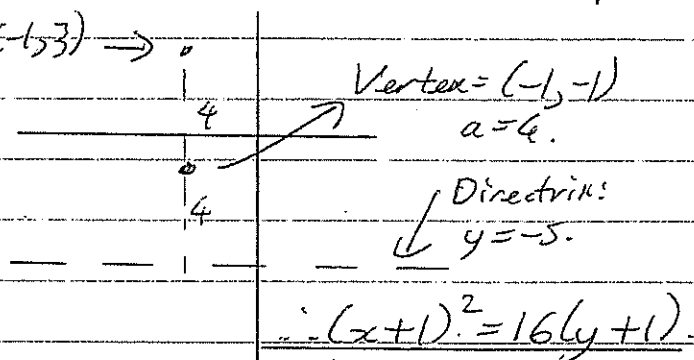
Q.(15)(b)(iii) [cont] - Point of inflexion
at $(-3, -\frac{2}{e^3})$

(iv)



Q.(16)(a) $\rightarrow$ Equidistant from point & line = parabola

$S = (-1, 3) \rightarrow$



(b)(i) $A = (x, x^2 + 2x + 5)$ $B = (x, 4x - x^2)$

Distance $AB = (x^2 + 2x + 5) - (4x - x^2)$
 $= 2x^2 - 2x + 5$

PTO $\rightarrow$

Q. (16) (b) (ii) Minimum AB:

$$\frac{d}{dx}(AB) = 4x - 2.$$

Finding minimum: $\frac{d}{dx}(AB) = 0$

$$4x - 2 = 0$$

$$x = \frac{1}{2}$$

Testing to see if minimum:

$$\frac{d^2}{dx^2}(AB) = 4 > 0$$

∴ Minimum AB at $x = \frac{1}{2}$.

$$\begin{aligned} \text{At } x = \frac{1}{2}, \quad AB &= 2x^2 - 2x + 15 \\ &= 2 \times \left(\frac{1}{2}\right)^2 - 2 \times \frac{1}{2} + 15 \\ &= \frac{9}{2} \text{ or } 4\frac{1}{2} \end{aligned}$$

∴ Minimum distance AB = $4\frac{1}{2}$ units.

$$(c)(i) 1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$$

$$a = 1, r = \frac{1}{3}$$

$$\text{By } S_n = \frac{a(1-r^n)}{1-r}$$

$$S_n = \frac{1(1-(\frac{1}{3})^n)}{1-\frac{1}{3}}$$

$$S_n = \frac{3}{2} [1 - (\frac{1}{3})^n]$$

PTO →

(16)(i)(a)(A)

$$1 + \frac{2}{3} + \frac{3}{9} + \frac{4}{27} + \frac{5}{81}$$

$$= 1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \frac{1}{81} \quad (1)$$

$$+ \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \frac{1}{81} \quad (2)$$

$$+ \frac{1}{9} + \frac{1}{27} + \frac{1}{81} \quad (3)$$

$$+ \frac{1}{27} + \frac{1}{81} \quad (4)$$

$$+ \frac{1}{81} \quad (5)$$

Using $S_n = \frac{a(1-r^n)}{1-r}$

$$(1) = \frac{1 \left[1 - \left(\frac{1}{3} \right)^5 \right]}{1 - \frac{1}{3}} = \frac{3}{2} - \frac{3}{2} \times \left(\frac{1}{3} \right)^5 = \frac{3}{2} - \frac{1}{2} \times \left(\frac{1}{3} \right)^4$$

$$(2) = \frac{\frac{1}{3} \left[1 - \left(\frac{1}{3} \right)^4 \right]}{1 - \frac{1}{3}} = \frac{1}{2} - \frac{1}{2} \times \left(\frac{1}{3} \right)^4 = \frac{1}{2} - \frac{1}{2} \times \left(\frac{1}{3} \right)^4$$

$$(3) = \frac{\frac{1}{9} \left[1 - \left(\frac{1}{3} \right)^3 \right]}{1 - \frac{1}{3}} = \frac{1}{6} - \frac{1}{6} \times \left(\frac{1}{3} \right)^3 = \frac{1}{6} - \frac{1}{2} \times \left(\frac{1}{3} \right)^4$$

$$(4) = \frac{\frac{1}{27} \left[1 - \left(\frac{1}{3} \right)^2 \right]}{1 - \frac{1}{3}} = \frac{1}{18} - \frac{1}{2} \times \left(\frac{1}{3} \right)^4$$

$$(5) = \frac{\frac{1}{81} \left[1 - \left(\frac{1}{3} \right) \right]}{1 - \frac{1}{3}} = \frac{1}{54} - \frac{1}{2} \times \left(\frac{1}{3} \right)^4$$

$$(1) + (2) + (3) + (4) + (5)$$

$$= \left(\frac{3}{2} + \frac{1}{2} + \frac{1}{6} + \frac{1}{18} + \frac{1}{54} \right) - \frac{5}{2} \times \left(\frac{1}{3} \right)^4$$

By $S_n = \frac{a(1-r^n)}{1-r}$

$$= \frac{\frac{3}{2} \left[1 - \left(\frac{1}{3} \right)^5 \right]}{1 - \frac{1}{3}} - \frac{5}{2} \times \left(\frac{1}{3} \right)^4$$

$$= \frac{9}{4} \left[1 - \left(\frac{1}{3} \right)^5 \right] - \frac{5}{2} \times \left(\frac{1}{3} \right)^4$$

= RHS QED.

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$$(16)(c)(ii)(B) S_5 = \frac{9}{4} \left[1 - \left(\frac{1}{3} \right)^5 \right] - \frac{5}{2} \times \left(\frac{1}{3} \right)^4$$

$$S_n = \frac{9}{4} \left[1 - \left(\frac{1}{3} \right)^n \right] - \frac{n}{2} \times \left(\frac{1}{3} \right)^{n-1}$$

$$(C) \text{ Limiting Sum: As } n \rightarrow \infty, \quad \left(\frac{1}{3} \right)^n \text{ \& } \left(\frac{1}{3} \right)^{n-1} \rightarrow 0$$

$$\therefore \text{ Limiting Sum} = \frac{9}{4} \times 1 \\ = \frac{9}{4}$$

Here endeth the solutions!!!